

Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102



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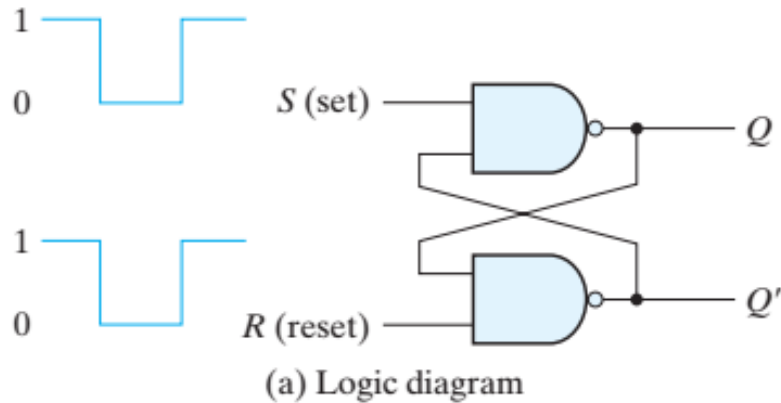
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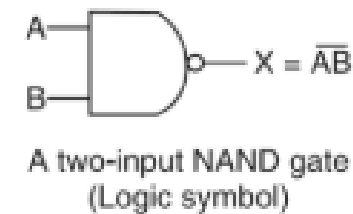
SR Latch

- ❖ The SR Latch is a circuit with two cross coupled NOR gates or two cross-coupled NAND gates, and two inputs labelled S for Set and R for Reset.
- ❖ The latch has two useful states. When $Q = 1$ and $\bar{Q} = 0$, then the Latch is said to be in the set state.
- ❖ When $Q = 0$ and $\bar{Q} = 1$, then the Latch is said to be in the reset state.

NAND SR Latch



S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

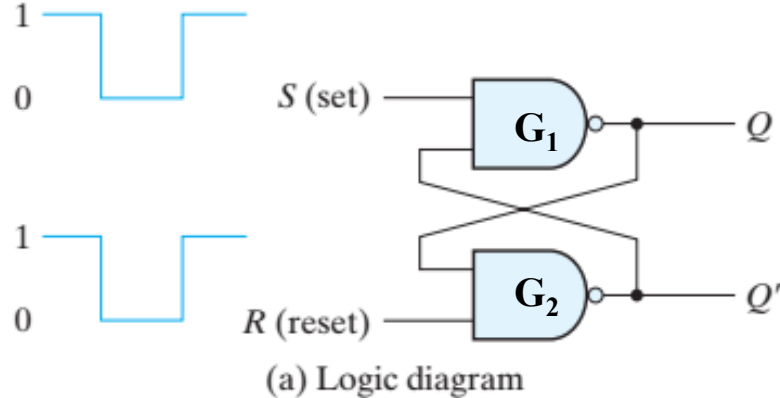


Truth table

Inputs		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

(b)

NAND SR Latch



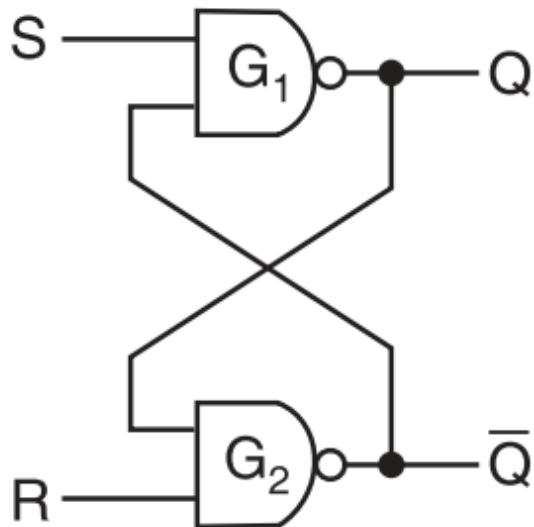
- ❖ In NAND gate if either of the input signal is 0 then the output is 1.
- ❖ Case 1: $S = 0, R = 1$ → The output of G_1 is 1 and the output of G_2 is 0.
- ❖ Case 2: $S = 1, R = 0$ → The output of G_2 is 1 and the output of G_1 is 0.
- ❖ Case 3: $S = 1, R = 1$ → The output of G_1 is Q and the output of G_2 is Q' .
- ❖ Case 4: $S = 0, R = 0$ → The output of G_1 is 1 and the output of G_2 is 1.

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

NAND SR Latch



S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

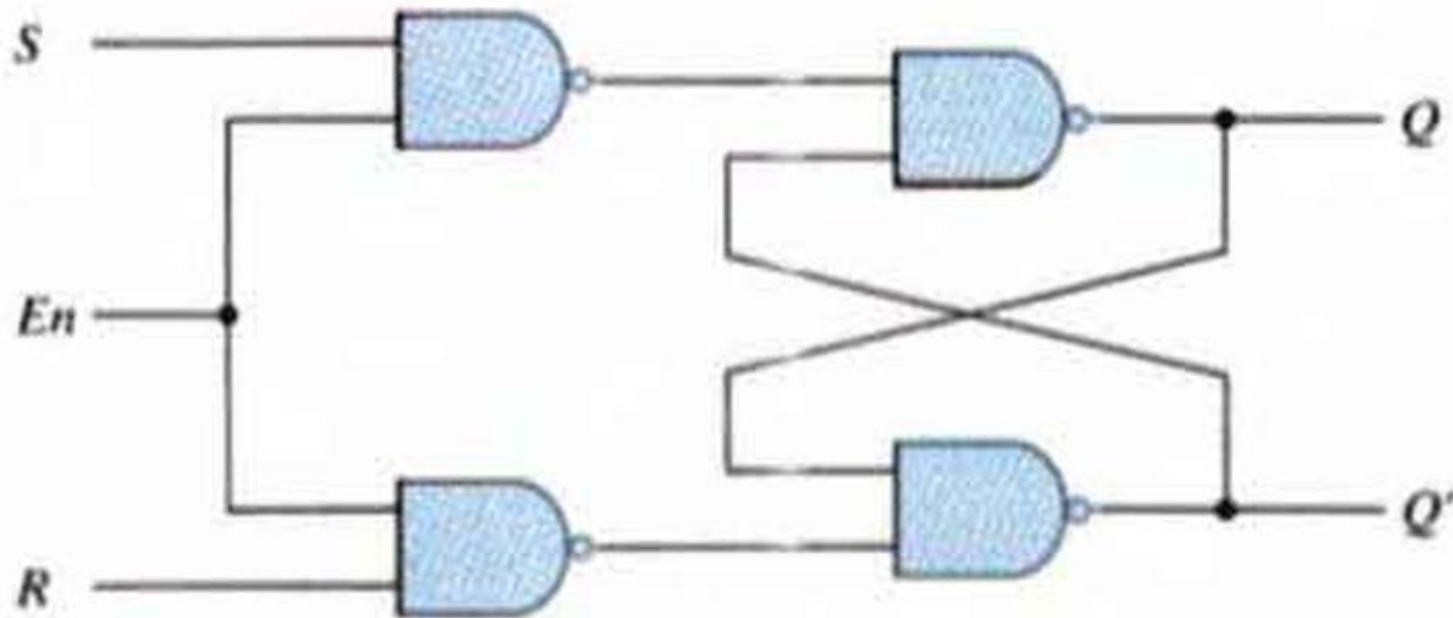


(a) Using NAND gates

S	R	Q_n	Q_{n+1}	State
0	0	0	×	Indeterminate (invalid)
0	0	1	×	
0	1	0	1	Set
0	1	1	1	
1	0	0	0	Reset
1	0	1	0	
1	1	0	0	No Change (NC)
1	1	1	1	

(b) Truth table

SR Latch with Control Input

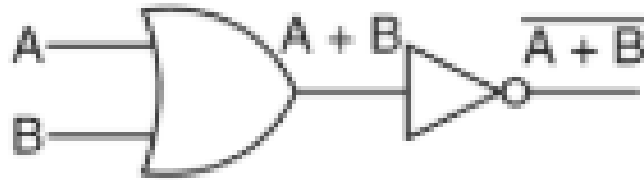


(a) Logic diagram

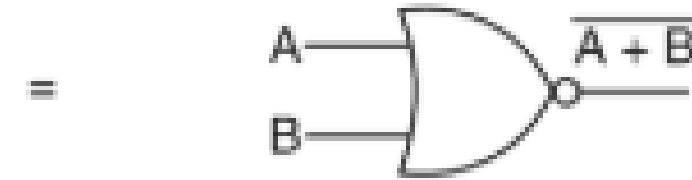
En	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

SR Latch with NOR Gates



An OR gate followed by a NOT gate

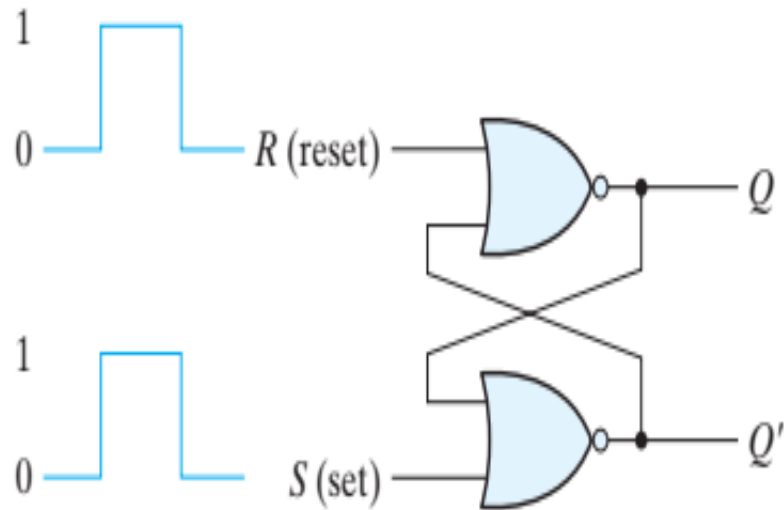


A two-input NOR gate (logic symbol)

(a)

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

(b) Truth table

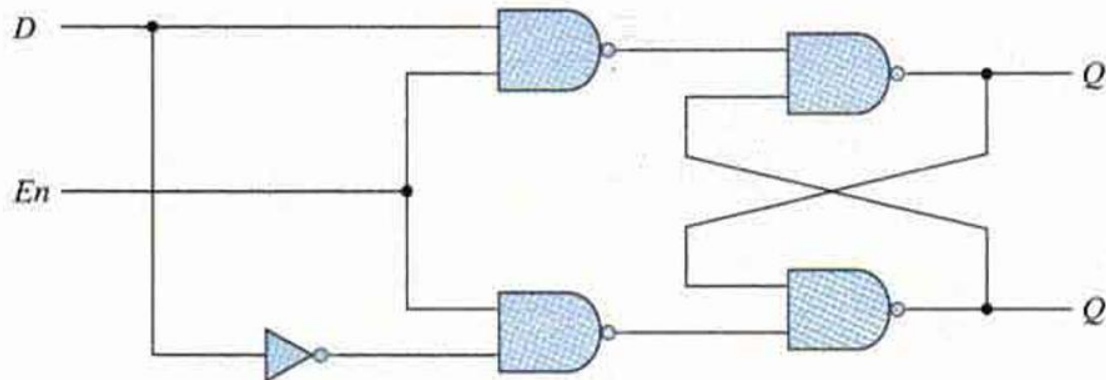


S	R	Q_{n+1}
0	0	No Change (Hold State)
0	1	0 (Reset)
1	0	1 (Set)
1	1	Invalid State

D Latch (Transparent Latch)



- ❖ One way to eliminate the undesirable condition of the indeterminate state in the SR latch is to ensure that inputs S and R are never equal to 1 at the same time. This is done in the D latch.
- ❖ **This latch has only two inputs: D (data) and En (enable).**
- ❖ **The D input goes directly to the S input and its complement is applied to the R input.**
- ❖ **As long as the enable input is at 0, the cross-coupled SR latch has both inputs at the 1 level and the circuit cannot change state regardless of the value of D.**
- ❖ The D input is sampled when $En = 1$.
- ❖ If $D = 1$, the Q output goes to 1, placing the circuit in the set state.
- ❖ If $D = 0$, output Q goes to 0, placing the circuit in the reset state.
- ❖ **The D latch receives that designation from its ability to hold data in its internal storage.**

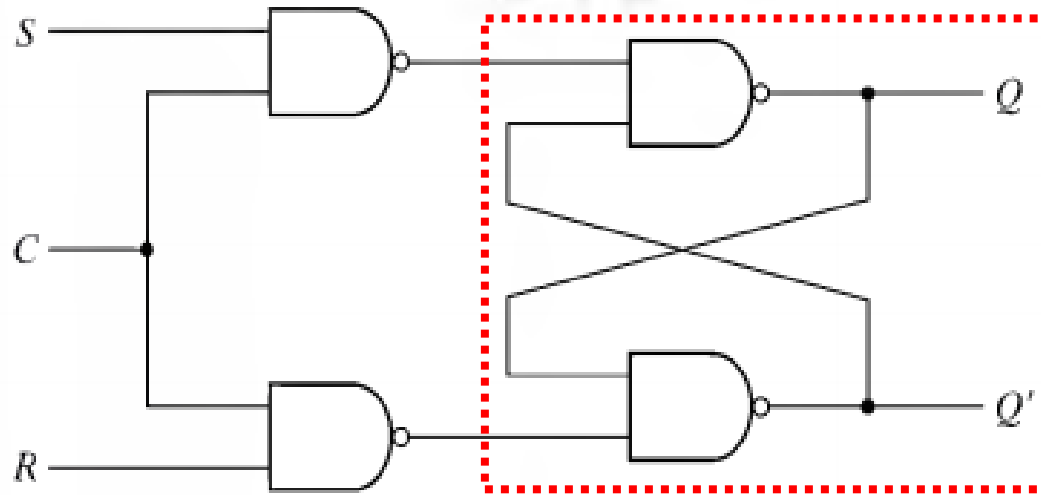


En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

(a) Logic diagram

(b) Function table

SR Flip Flop using NAND Gate



Truth Table of SR Latch

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

Truth Table of SR Flip Flop

CLK	S	R	Q_{n+1}
0 (Not Triggered)	x	x	No Change (Hold State)
1 (Triggered)	0	0	No Change (Hold State)
1 (Triggered)	0	1	0 (Reset)
1 (Triggered)	1	0	1 (Set)
1 (Triggered)	1	1	Invalid State

SR Flip Flop Characteristic and Excitation Table

Truth Table of SR Flip Flop

CLK	S	R	Q_{n+1}
0 (Not Triggered)	x	x	No Change (Hold State)
1 (Triggered)	0	0	No Change (Hold State)
1 (Triggered)	0	1	0 (Reset)
1 (Triggered)	1	0	1 (Set)
1 (Triggered)	1	1	Invalid State

Characteristic Table

- Determined by truth table.
- Next state depends upon input (S,R) and also on previous state.

Characteristics Table of SR Flip Flop

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

Excitation Table

Determined by characteristic table

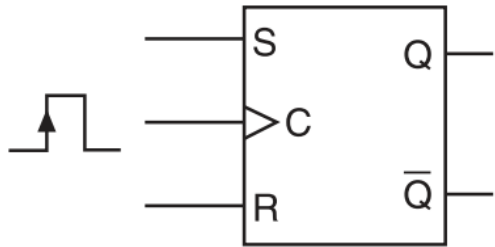
Excitation Table of SR Flip Flop

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

$$Q_{n+1} = S + Q_n \bar{R}$$

$S \backslash RQ_n$	00	01	11	10
0	0	1	0	0
1	1	1	x	x

Positive Edge Triggered Flip Flop



(a) Logic symbol

C	S	R	Q_n	Q_{n+1}	State
↑	0	0	0	0	No Change (NC)
↑	0	0	1	1	
↑	0	1	0	0	Reset
↑	0	1	1	0	
↑	1	0	0	1	Set
↑	1	0	1	1	
↑	1	1	0	x	Indeterminate
↑	1	1	1	x	
0	x	x	0	0	No Change (NC)
0	x	x	1	1	

(b) Truth table

Truth Table of SR Flip Flop

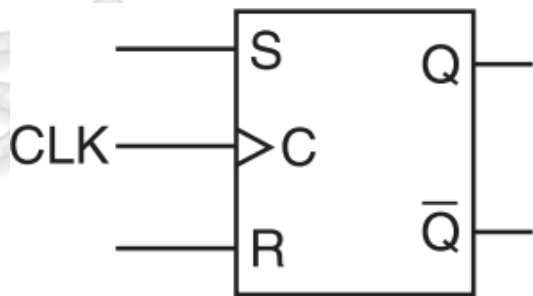
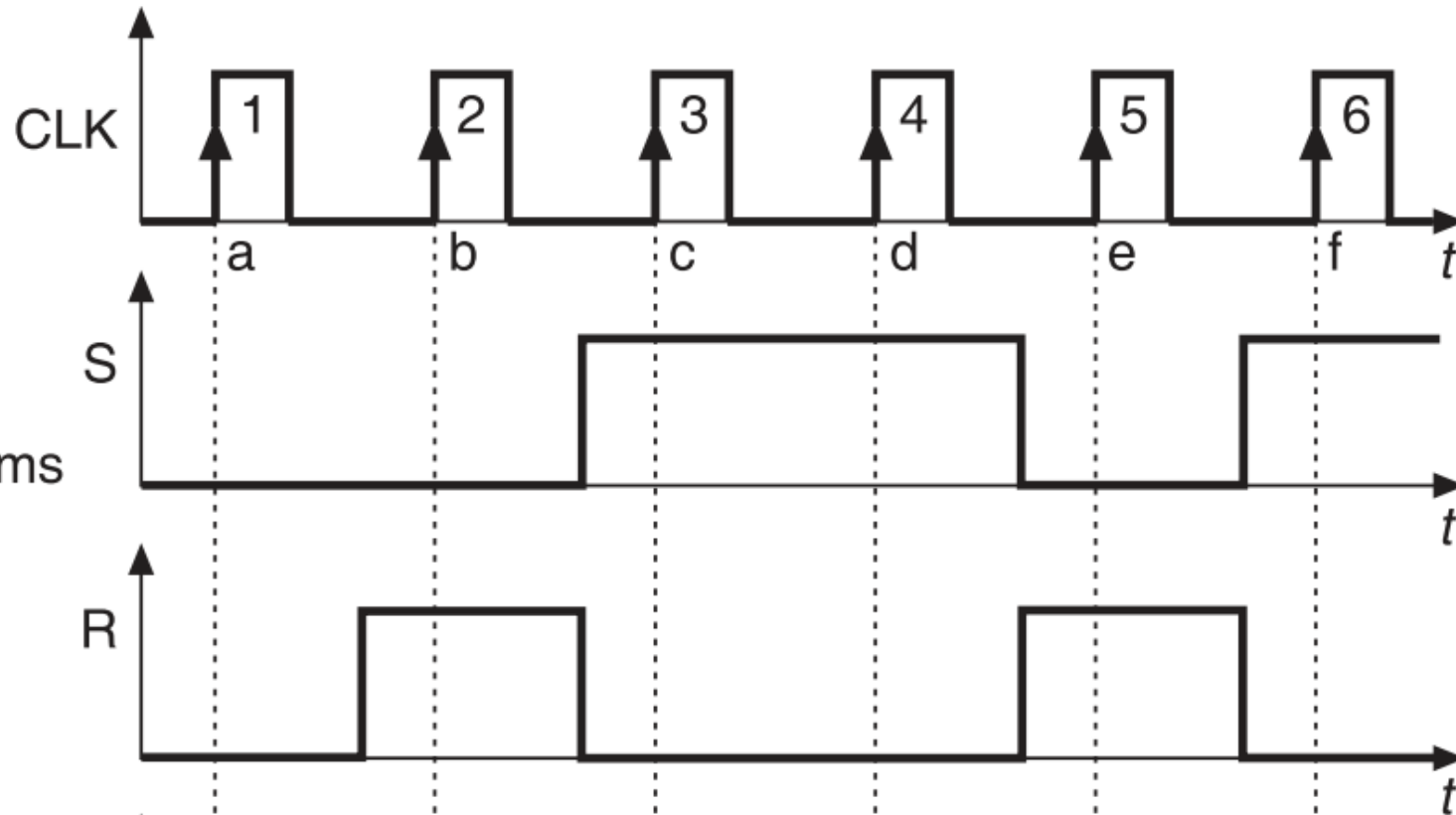
CLK	S	R	Q_{n+1}
0 (Not Triggered)	x	x	No Change (Hold State)
1 (Triggered)	0	0	No Change (Hold State)
1 (Triggered)	0	1	0 (Reset)
1 (Triggered)	1	0	1 (Set)
1 (Triggered)	1	1	Invalid State

Positive Edge Triggered Flip Flop

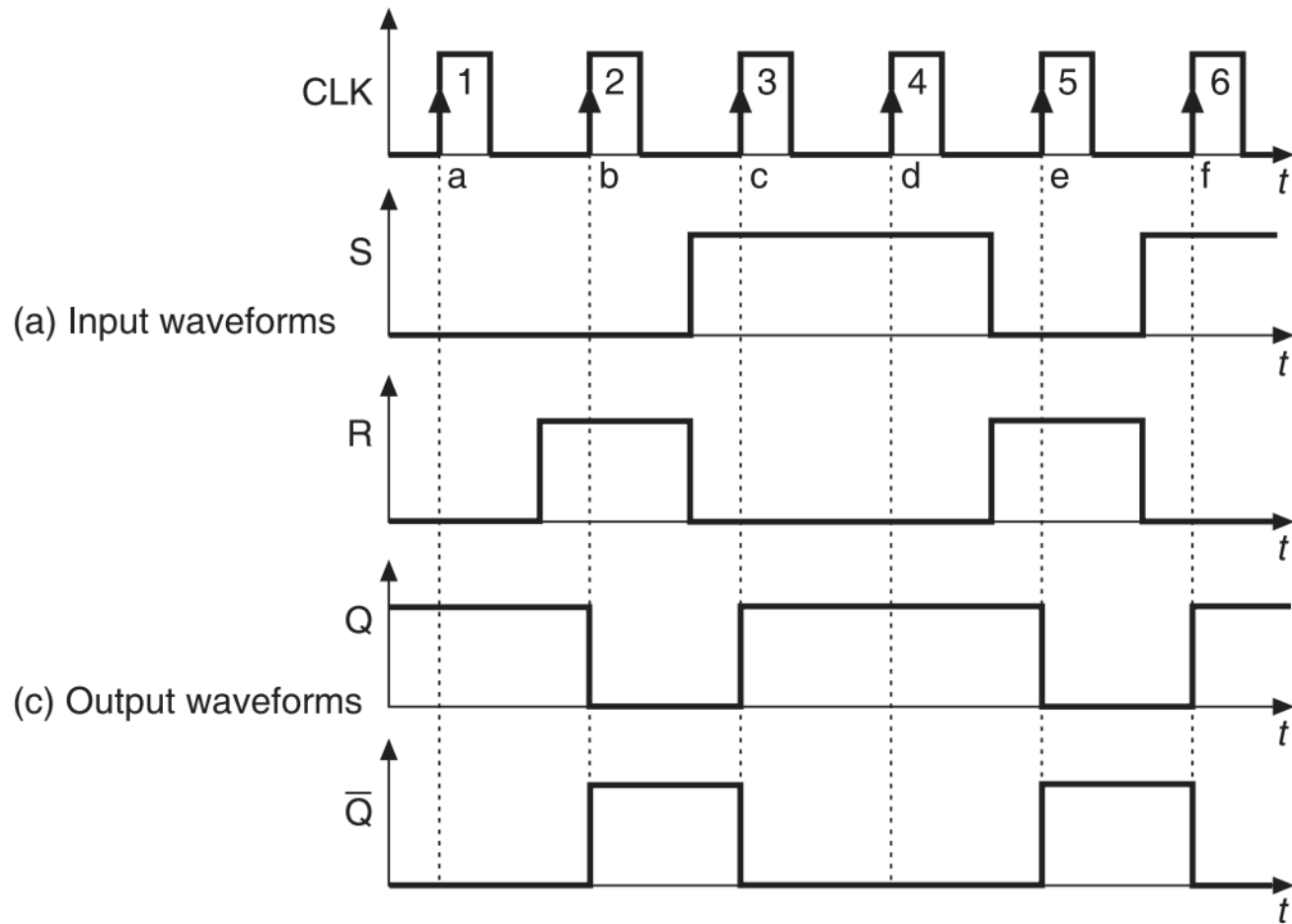


Problem: The waveforms shown in Figure a are applied to the positive edge-triggered S-R flip-flop shown in Figure b. Sketch the output waveforms.

(a) Input waveforms



Positive Edge Triggered Flip Flop



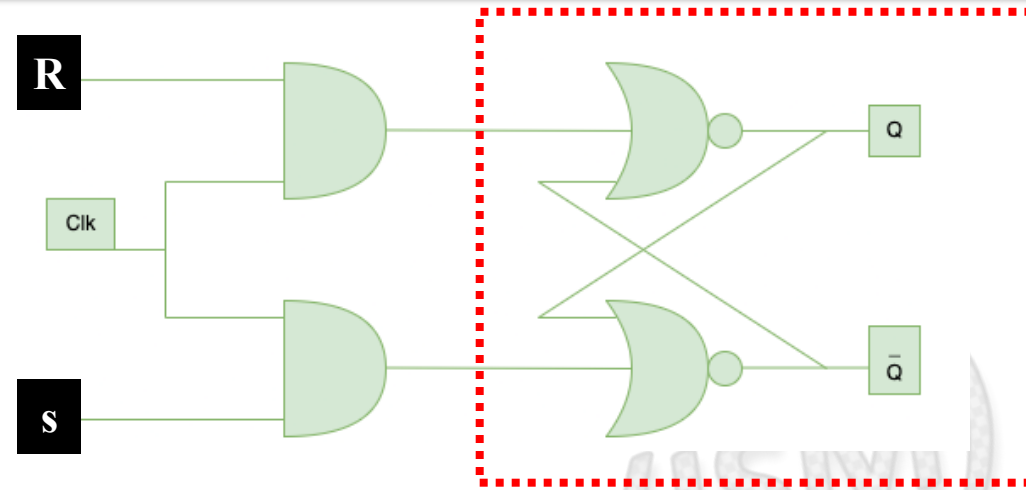
CLK	S	R	Q_{n+1}
0 (Not Triggered)	x	x	No Change (Hold State)
1 (Triggered)	0	0	No Change (Hold State)
1 (Triggered)	0	1	0 (Reset)
1 (Triggered)	1	0	1 (Set)
1 (Triggered)	1	1	Invalid State

Positive Edge Triggered Flip Flop



1. Initially, $S = 0$ and $R = 0$ and Q is assumed to be HIGH.
2. At the positive-going transition of the first clock pulse (i.e. at a), both S and R are LOW.
So, no change of state takes place. Q remains HIGH and $\bar{Q}$ remains LOW.
3. At the leading edge of the second clock pulse (i.e. at b), $S = 0$ and $R = 1$. So, the flip-flop resets. Hence, Q goes LOW and $\bar{Q}$ goes HIGH.
4. At the positive-going edge of the third clock pulse (i.e. at c), $S = 1$ and $R = 0$. So, the flip-flop sets. Hence, Q goes HIGH and $\bar{Q}$ goes LOW.
5. At the rising edge of the fourth clock pulse, $S = 1$ and $R = 0$. Since the flip-flop is already in a SET state, it remains SET. That is, Q remains HIGH and $\bar{Q}$ remains LOW.
6. The fifth pulse resets the flip-flop at its positive-going edge because $S = 0$ and $R = 1$ is the input condition and $Q = 1$ at that time.
7. The sixth pulse sets the flip-flop at its rising edge because $S = 1$ and $R = 0$ is the input condition and $Q = 0$ at that time.

SR Flip Flop using NOR Gate



Truth Table of SR Flip Flop

CLK	S	R	Q_{n+1}
0 (Not Triggered)	x	x	No Change (Hold State)
1 (Triggered)	0	0	No Change (Hold State)
1 (Triggered)	0	1	0 (Reset)
1 (Triggered)	1	0	1 (Set)
1 (Triggered)	1	1	Invalid State